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Btech CSE core
Sec-A

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Topic Date

Assignment :- 1 DBMS Unit - 1

Sec-A Introduction to DBMS-

Q-1. DBMS - A DBMS is software that manages data in an organised way using tables. It helps users to store, update & retrieve data easily.
It is required to manage large data safely, avoid duplication, & provide secure access.

2. Why DBMS is better than file system?
DBMS stores data in a structured format & allows multiple users to access it safely.
File systems may create duplicate data & are harder to manage.

3- Data Abstraction- It means hiding technical detail & showing only required information to user.
- Physical (how data is stored).
- Logical (what data is stored)
- View (what user sees)

4- What is Three-Schema Architecture?
It divides database into three parts-
- Internal level
- Conceptual level
- External level
It helps in data independence & easy modification.

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5- Data Independence - It means changing dataset structure at one level without affecting other levels.
It helps in flexibility & System maintenance.

6- ER model - It is a diagram that represents entities, their attributes, & relationships. It is used for database design planning.

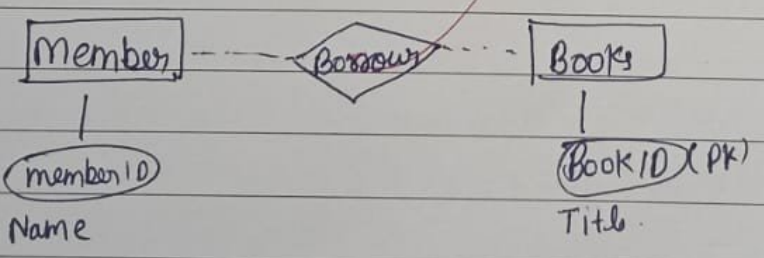
7- What are Integrity Constraints - Integrity constraints are rules that ensure correct & valid data.
Ex- Primary keys, Foreign key, Not null.

Sec-B- Design, Draw & Defend-

1- Three - Schema-

External level → user view
↓
Conceptual level → overall structure
↓
Internal level → physical storage

2. Simple ER diagram → Library



- Entities: Member, Book
- Relationship - Borrow
- PK - member ID, Book ID

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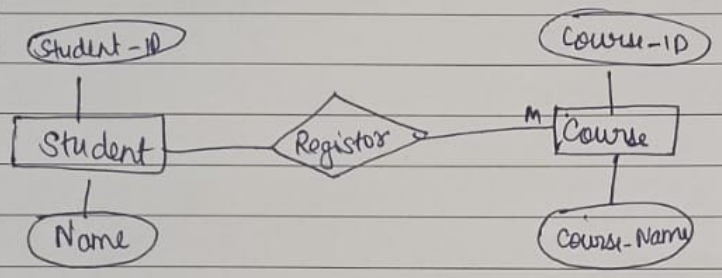
One member can borrow many books

(iii) Student - Course Registration ER

- Student : (Student ID, Name)
- Course (Course-ID, Course-Name).

Relationship : Student registor for Course.



Sec-C- Case Based Analytical Ques-

(i) ER Diagram- Hospital System

Entities - Patient, Doctor, Appointment
Patients books Appointment.
Doctor attends Appointment.

Primary keys - Patient-ID, Doctor-ID, Appointment-ID.

(ii) Online Shopping-
Customer. (customer ID - PK)
Product (Product-ID - PK)

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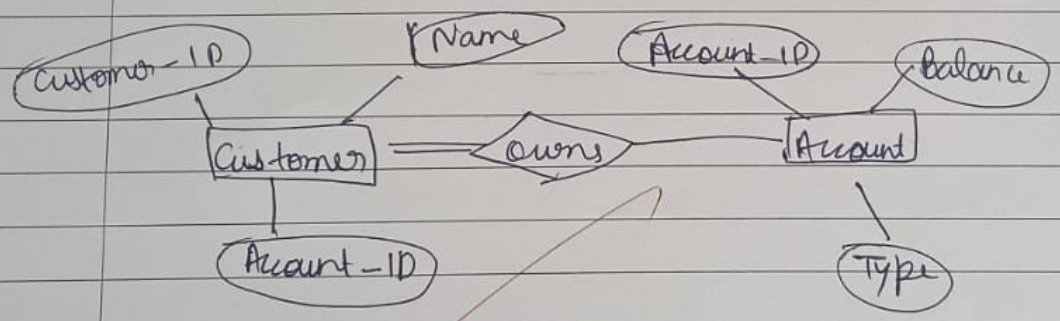
Order (Order-ID-PK)
foreign Key in order Table connect Customer & Product.

(iii) University - Data Abstraction-
Student see marks (view level).
Admin manages records (logical level)
DBA . Controls Storage (Physical level)

Sec-D- One diagram + On Insign-

(i) ER-Diagram - ATM System-



Account-no is Primary key
Relation : Customer owns Account.
Account-no take uniquely Indentifier Account.

(ii) Three - Schema Architecture + one Benifit-

Benefit- Provide Data Independence.
Change in storage do not affect uses.

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Diagram:- External level (user view)
↓
Conceptual level (logical structure of Database)
↓
Internal level (Physical storage level).

(iii) Draw a Entity with Attributes.

Entity: Employee.
Employee - ID (PK)
Name
Salary.
Department

Primary key - Employee - ID.
because for all Employee their ID will be different.

Section - E. Reflection, Assumption & Viva.

1. Two assumption while drawing ER-diagram.
Ans. Each student has a unique student-ID.
- One appointment is linked to only one doctor.

2. One limit of ER modelling.
Ans. Cannot represent complex Constraints Easily
- Does not show procedural aspects
- Not suitable for of large dynamic system directly.

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(iii) Most Difficult Part in Unit-1

Ans. The most difficult part was understanding data independence, because it requires clear understanding of three-Schema architecture & how level interact.